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XMODEL, GLISTER & MODELZEN

Setup Guide

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Outlines

- This document explains how to:
 - Install the *XMODEL* package
 - Setup for *XMODEL*
 - Setup for *GLISTER*
 - Additional Setup for *MODELZEN*

XMODEL, GLISTER, and MODELZEN



XMODEL

- ***XMODEL*** enables fast and accurate simulation of analog/mixed-signal systems entirely in SystemVerilog



GLISTER

- ***GLISTER*** is a graphical user interface of XMODEL and MODELZEN integrated into Cadence Virtuoso Schematic Editor



MODELZEN

- ***MODELZEN*** can automatically generate XMODEL models from your circuit schematics or netlists

Installing *XMODEL* Package

- First, download the *XMODEL* installation package from Scientific Analog's website
 - `xmodel_YYYY.MM_x86_64.sh` for Linux 64-bit platforms
 - `xmodel_YYYY.MM_x86.sh` for Linux 32-bit platforms
 - `YYYY.MM` is the release year and month
- Second, make the file executable and execute it:

```
$ chmod +x xmodel_2025.02_x86_64.sh  
$ ./xmodel_2025.02_x86_64.sh
```

 - And follow the instructions as prompted

Installing *XMODEL* Package (2)

- The one-step way of installing *XMODEL* is to use **-b** and **-p** options:

```
$ chmod +x xmodel_2025.02_x86_64.sh  
$ ./xmodel_2025.02_x86_64.sh -b -p <INSTALL_PATH>
```

- **-b** option: runs in batch mode without prompting
- **-p** option: specifies the installation path <INSTALL_PATH>

Environment Setup for *XMODEL*

- To use *XMODEL*, you need to define the following environment variables:
 - **\$XMODEL_HOME**
: the *XMODEL* installation path
 - **\$XMODEL_SIMULATOR**
: default SystemVerilog simulator (vcs, ncverilog, or modelsim)
 - **\$SALMD_LICENSE_FILE**
: FlexNet license server hostname or file path
 - **\$PATH**
: append **\$XMODEL_HOME/bin** to existing \$PATH

Environment Setup for *XMODEL* (2)

- Examples

- In bash-like shells:

```
export XMODEL_HOME=/cad/xmodel/xmodel_2025.02
export XMODEL_SIMULATOR=vcs
export PATH=${PATH}:${XMODEL_HOME}/bin
export SALMD_LICENSE_FILE=9524@licserver
```

- In csh-like shells:

```
setenv XMODEL_HOME /cad/xmodel/xmodel_2025.02
setenv XMODEL_SIMULATOR vcs
setenv PATH ${PATH}:${XMODEL_HOME}/bin
setenv SALMD_LICENSE_FILE 9524@licserver
```

* NOTE: please refer to other sources on the differences between UNIX shells

Install/Setup SystemVerilog Simulator

- *XMODEL* can work with Synopsys VCS, Cadence NC-Verilog (i.e. INCISIVE), and MentorGraphics ModelSim (QuestaSim)
 - Please refer to the corresponding vendor's manuals on how to install and setup each simulator
- One example: ModelSim Altera-Starter Edition (ASE)
 - Freely available (with some limitations on speed/capacity)
 - Download site (requires user registration):
http://dl.altera.com/?edition=standard&product=modelsim_ae#tabs-2

ModelSim ASE Setup Example

- To setup ModelSim, define **\$MODELSIM_HOME** and add **\$MODELSIM_HOME/bin** to **\$PATH**

```
export MODELSIM_HOME=/cad/modelsim-ASE-16.0/modelsim_ase
export PATH=${PATH}:${MODELSIM_HOME}/bin
```

- And for *XMODEL*, use "modelsim" for **\$XMODEL_SIMULATOR**:

```
export XMODEL_HOME=/cad/xmodel/xmodel_2025.02
export XMODEL_SIMULATOR=modelsim
export PATH=${PATH}:${XMODEL_HOME}/bin
export SALMD_LICENSE_FILE=9524@licserver
```

XMODEL Package Organization

- **/doc:** contains documentation files
- **/bin:** contains executable programs (*xmodel*, *xwave*, ...)
- **/lib:** contains platform-specific run-time libraries
- **/etc:** default configurations, model templates, etc.
- **/function, /stim, /gate, /vdt, /connect, /meas, /circuit:** contains the XMODEL primitive modules
- **/cadence:** contains Cadence-related libraries and setup files

XMODEL Package Organization (2)

- **/block**: contains block-level modeling examples (phase-frequency detector, voltage-controlled oscillators, etc).
- **/example**: contains system-level modeling examples (charge-pump PLL, bang-bang CDR, high-speed links, adaptive DFE equalizer, etc).
- **/tutorial**: contains tutorial materials on *XMODEL*, *GLISTER*, and *MODELZEN* basics and high-speed links.

About XMODEL Examples

- Examples in **/example** directory showcase various analog/mixed-signal systems modeled using *XMODEL*
- Most models are written in text and simulated via Makefile script (soon to be replaced with *MODELBOX*)
- Examples include:
 - ***cp_pll***, ***dpll***: charge pump PLL and digitally-controlled PLL
 - ***bb_cdr***: bangbang controlled CDR
 - ***eq_adapt***: equalizer adaptation
 - ***hs_link***: high-speed link
- Refer to the PDF documentation contained in each sub-directory for further information

Running XMODEL Examples

- Copy the example directory to your local directory:

```
$ cp -R $XMODEL_HOME/example ~/example
```

- Change directory into the digital PLL example (*dpll*):

```
$ cd ~/example/dpll
```

- Change directory into the testbench simulating the locking transient of the PLL (*tb_locking*):

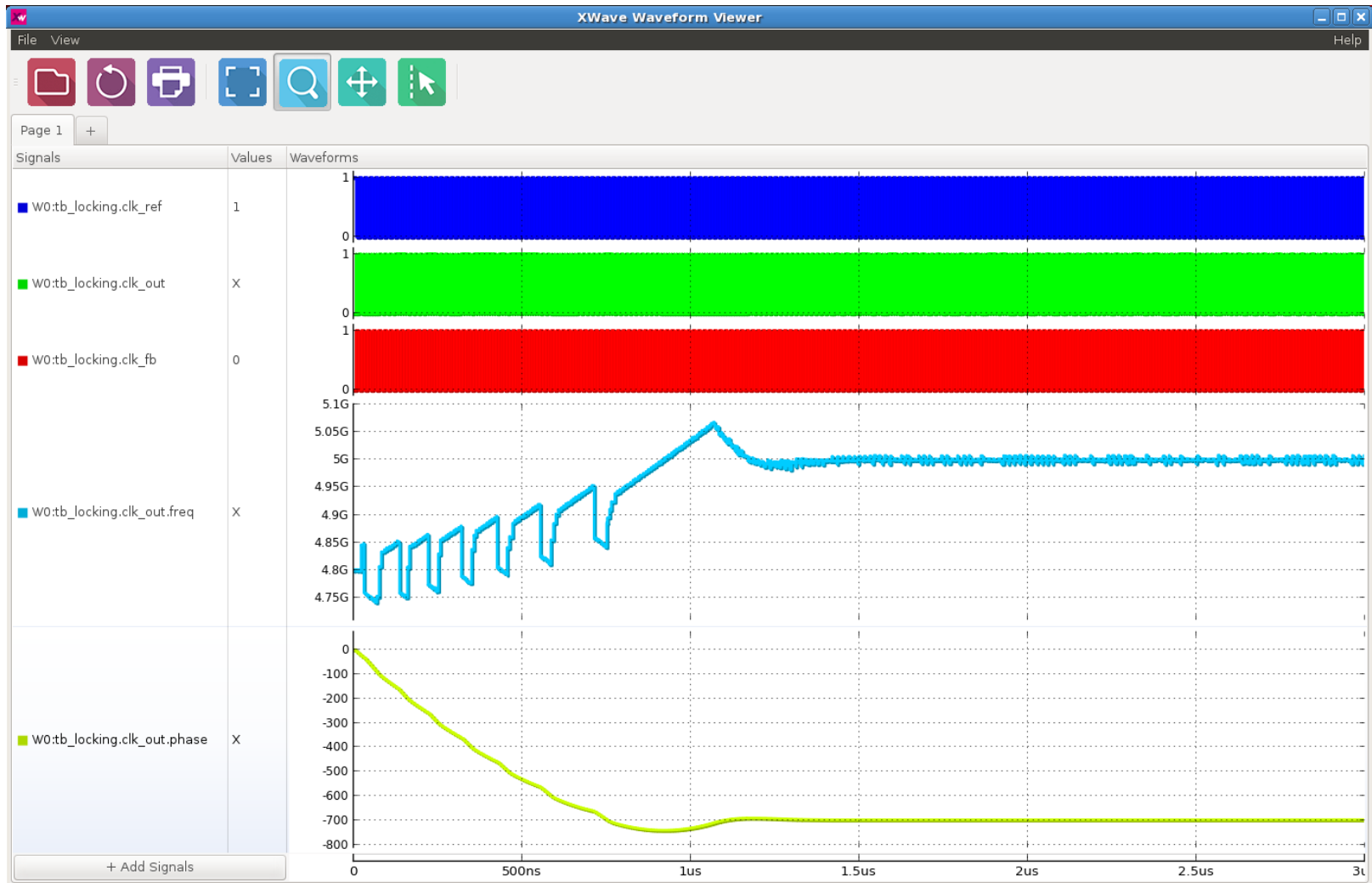
```
$ cd tb/tb_locking
```

- Run simulation and plot waveforms using XWAVE:

```
$ make  
$ xwave -a xmodel.jez
```

Find documentations on this example in:
[\\$XMODEL_HOME/doc/Example_DPLL.pdf](#)

Simulation Results



XMODEL Tutorial

- The *XMODEL* tutorial included in the *XMODEL* Package covers how to use *XMODEL* in the UNIX command-line interface
 - Located in **`$XMODEL_HOME/tutorial/xmodel_basic`**
 - Follow the instructions described in README file to get started
 - The **`doc/`** sub-directory contains the full tutorial materials

Virtuoso Setup for *GLISTER*

- Install the *XMODEL* registry files to the Cadence installation directory by executing:

```
$XMODEL_HOME/bin/setup_registry --cdshome $CDSHOME
```

- **\$CDSHOME** is the Cadence installation directory
- Alternatively, you can install the *XMODEL* registry files to a local directory reachable by the CSF mechanism:
 - e.g. the work directory where you launch Virtuoso from or the directory defined by **\$CDS_PROJECT** or **\$CDS_SITE**
 - The command is:

```
cd $CDS_PROJECT or <your cadence work directory>  
$XMODEL_HOME/bin/setup_registry --local .
```

Virtuoso Setup for *GLISTER* (2)

- Insert the following lines to your **.cdsinit** file

```
printf("Loading XMODEL/Virtuoso Integration...\n")
XMODEL_HOME = getShellEnvVar("XMODEL_HOME")
loadContext( strcat( XMODEL_HOME "/cadence/etc/xmodel.cxt" ) )
callInitProc( "xmodel" )
```

- And insert these lines to **cds.lib** file

```
INCLUDE ${XMODEL_HOME}/cadence/etc/cds.lib
```

or:

```
DEFINE xmodel_prims      ${XMODEL_HOME}/cadence/xmodel_prims
DEFINE xmodel_blocks     ${XMODEL_HOME}/cadence/xmodel_blocks
```

UNIX Setup for *GLISTER*

- Define **\$XMODEL_SIMDIR** environment variable to set where the model files are stored after netlisting
 - Default is **\$CDS_PROJECT/xmodel.sim** if not defined

- Example:

- For bash-like shells:

```
export XMODEL_SIMDIR=$CDS_PROJECT/xmodel.sim
```

- For csh-like shells:

```
setenv XMODEL_SIMDIR $CDS_PROJECT/xmodel.sim
```

GLISTER Tutorials

- The XMODEL package includes two tutorials on how to use *XMODEL* within Cadence Virtuoso Environment
 - **\$XMODEL_HOME/tutorial/glister_basic**: overview on how to compose analog models with *XMODEL* primitives and run simulations, using a SAR-ADC as an example
 - **\$XMODEL_HOME/tutorial/xmodel_hslink**: overview on how to model various components of high-speed I/O interface and simulate its system-level performances using *XMODEL*
- Follow the instructions described in the README files to get started
- The **doc/** sub-directory contains the full tutorial slides

MODELZEN Setup

- To use *MODELZEN*, you need to setup a technology configuration file for your circuit's technology
- Also, depending on your design environment, you may need to do some additional Virtuoso setups:
 - Override *MODELZEN*-related *GLISTER* options if necessary
 - Redefine how *MODELZEN* extracts circuit netlists from schematic cellviews (***modelzenGenNetlist()***)

MODELZEN Technology Configuration

- *MODELZEN* technology configuration file defines the technology-specific mapping and environment options necessary to convert a circuit netlist to XMODEL/SystemVerilog model
 - A sample in `$XMODEL_HOME/etc/tech_config.py`
- Its key sections are:
 - SPICE simulator commands and options
 - SPICE device model libraries and corners
 - Device mapping between SPICE and XMODEL
- For a starter, refer to the *MODELZEN Setup Guide*

Virtuoso Setup for *MODELZEN*

- You can override the *MODELZEN*-related *GLISTER* options by redefining some SKILL variables after loading the *XMODEL* context in the *.cdsinit* file:

```
; XMODEL/Virtuoso Integration Setup
printf( "Loading XMODEL/Virtuoso Integration...\n" )
XMODEL_HOME = getShellEnvVar("XMODEL_HOME")
loadContext( strcat( XMODEL_HOME "/cadence/etc/xmodel.cxt" ) )
callInitProc( "xmodel" )

; MODELZEN options (NOTE: must come after context file loading)
modelzenDefaultModelView = "xmodel"
modelzenDefaultConfigFile = "./tech_config.py"
modelzenDefaultRunDir = "./modelzen.run"
modelzenDefaultNetlistSimulator = "hspiceD"
modelzenSubModulePrefix = "SUB_%CELLNAME%_"
modelzenShowOptionDialog = t
```

MODELZEN Option Variables

- The list of a few selected options that control the way *MODELZEN* is run from Virtuoso:

Option Variable Name	Description	Default Value
<code>modelzenDefaultModelView</code>	Default viewname of the model generated by <i>MODELZEN</i>	"xmodel"
<code>modelzenDefaultConfigFile</code>	Default technology configuration file used by <i>MODELZEN</i>	"./tech_config.py"
<code>modelzenDefaultRunDir</code>	Directory to store <i>MODELZEN</i> results (/netlist, /sim, /model)	"./modelzen.run"
<code>modelzenDefaultNetlistSimulator</code>	Cadence netlist simulator name (e.g. hspiceD, spectre, cd1, etc.)	"hspiceD"
<code>modelzenSubModulePrefix</code>	Optional prefix added to the submodules included in the model	"SUB_%CELLNAME%_"
<code>modelzenShowOptionDialog</code>	Flag whether to show the <i>MODELZEN</i> GUI dialog for user's confirmation	t

Schematic Netlisting for *MODELZEN*

- You can customize the way *MODELZEN* extracts SPICE/Spectre netlists from circuit schematics by overriding ***modelzenGenNetlist()*** function
- The format of ***modelzenGenNetlist()*** is:

```
modelzenGenNetlist( libName cellName viewName simulator )
```

- ***libName, cellName, viewName***: library, cell, and view names of the design to be netlisted
- ***simulator***: netlist simulator format (e.g. hspiceD, spectre, cdl, etc.)
- Returns the full path to the netlist file

Example of *modelzenGenNetlist()*

- Generating netlists using Cadence ADE netlister:

```
; modelzenGenNetlist( libName cellName viewName simulator )
;   generates the 'simulator' netlist of a cellview 'libName':'cellName':'viewName'
;   returns the filename of the generated netlist
procedure( modelzenGenNetlist( libName cellName viewName simulator )
  let( ( netlistFile netlistDir )
    ; initialize simulator
    simulator( simulator )

    ; generate netlist
    design( libName cellName viewName )
    unless( netlistFile = createNetlist( ?recreateAll t ?display nil )
      error( "error occurred while netlisting %s.%s:%s." libName cellName viewName )
    )
    ocnCloseSession()

    ; return netlist filename
    netlistDir = xmodelGetDirName( netlistFile )
    strcat( netlistDir "/netlist" )
  )
)
```

MODELZEN Tutorial

- The *MODELZEN* tutorial included in the *XMODEL* Package covers how to use *MODELZEN* to generate models from circuits
 - Located in **\$XMODEL_HOME/tutorial/modelzen_basic**
 - Follow the instructions described in README file to get started
 - The **doc/** sub-directory contains the full tutorial materials